CS 1671 / CS 2071 / ISSP 2071
Human Language Technologies

Session 25: Information retrieval, RAG

Michael Miller Yoder

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University of Pittsburgh
School of Computing and Information

Assessments: homework

- Homework 3 is **due tomorrow, Thu Apr 16**
- Run Jupyter notebooks from templates on the CRCD
- Part 1: LLM prompting
  - Use CRCD CPUs
- Part 2: Instruction tuning of an LLM
  - Use CRCD GPUs

Assessments: project

- Project final report is **due Tue Apr 28**
- Implement your proposed LLM system, compare with your baseline system
- See project website for instructions of what to include
- Project presentation is **in class Fri May 1, 10-11:50am**
  - Ungraded
  - Michael will post a shared PowerPoint

Assessments: extra credit quiz

- Optional, extra credit quiz during class **this Mon Apr 20** covering:
  - Session 21: Prompting, finetuning, and evaluating LLMs
    - J+M 7.6, 8.8.3, 9.4
  - Session 24: Machine translation
    - J+M 12-12.4, 12.6-12.8
  - Session 25 (today): Information retrieval, RAG
    - J+M 11

Learning objectives: information retrieval (IR), RAG

Students will be able to:

- Diagram the process of **classic information retrieval based on sparse embeddings**
- Explain the goal of word frequency weighting schemes like **tf-idf**
- Describe how **retrieval-augmented generation (RAG)** works
- List software that can be used to build classic IR systems and RAG
- Identify and explain a common evaluation IR evaluation metric, **mean reciprocal rank (MRR)**

Information retrieval and question answering

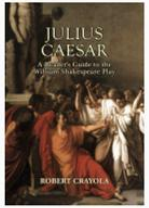
- Information retrieval (IR)
  - Choosing the most relevant document/s from a set of documents given a user's query
  - Search engines
- Closely related to question answering (QA)

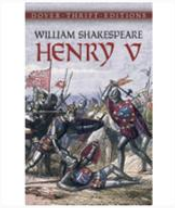
Traditional IR with tf-idf and BM25 weighting

Traditional IR: sparse embeddings

Sparse embeddings (bag-of-words) of documents and queries

- Each cell is the count of term t in a document d ($tf_{t,d}$).
- Each document is a count vector in $\mathbb{N}^V$, a column below.





	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	1	8	15
soldier	2	2	12	36
fool	37	58	1	5
clown	6	117	0	0

Slide based on David Mortensen, Jurafsky & Martin

Raw frequency is a bad representation

- The co-occurrence matrices we have seen represent each cell by word frequencies
  - Whether in term-document or term-term matrices
- Frequency is clearly useful; if *sugar* appears a lot near *apricot*, that's useful information.
- But overly frequent words like *the*, *it*, or *they* are not very informative about the context
  - 2 documents that use a lot of *the* are not necessarily similar
- It's a paradox! How can we balance these two conflicting constraints?

Slide adapted from Jurafsky & Martin

Weighting words in term-document and term-term matrices

- **tf-idf** (term frequency-inverse document frequency)
  - For representing documents with their most unique words (for text classification, information retrieval)
  - Term-document matrix

Slide adapted from Jurafsky & Martin

Term frequency (tf)

$tf_{t,d} = count(t,d)$

Instead of using raw count, we squash a bit:

$$tf_{t,d} = \begin{cases} 1 + \log count(t,d) & \text{if } count(t,d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

Slide adapted from Jurafsky & Martin

Document frequency (df)

df_t is the number of documents term *t* occurs in.

(note this is not collection frequency, which is the total count across all documents)

"Romeo" is very distinctive for one Shakespeare play:

	Collection Frequency	Document Frequency
Romeo	113	1
action	113	31

Slide adapted from Jurafsky & Martin

Inverse document frequency (idf)

$$\text{idf}_t = \log_{10}\left(\frac{N}{\text{df}_t}\right)$$

- N is the total number of documents in the collection
- Documents can be whatever you want! (Full documents, paragraphs, etc)

Word	df	idf
Romeo	1	1.57
salad	2	1.27
Falstaff	4	0.967
forest	12	0.489
battle	21	0.246
wit	34	0.037
fool	36	0.012
good	37	0
sweet	37	0

Slide adapted from Jurafsky & Martin

tf-idf Controls for Frequent but Uninformative Words

Some words are very common in a given document because they are common across all documents (e.g., *the*). They are not discriminative. `tf-idf` (product of term frequency and inverse document frequency) addresses this:

$$tf_{t,d} = \begin{cases} 1 + \log count(t,d) & \text{if } count(t,d) > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\text{idf}_f = \log_{10} \frac{N}{\text{df}_t}$$

$$\text{tf-idf}(t, d) = \text{tf}_{t,d} \cdot \text{idf}_t$$

Slide adapted from Jurafsky & Martin, David Mortensen

Final tf-idf weighted values

Raw counts

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

tf-idf:

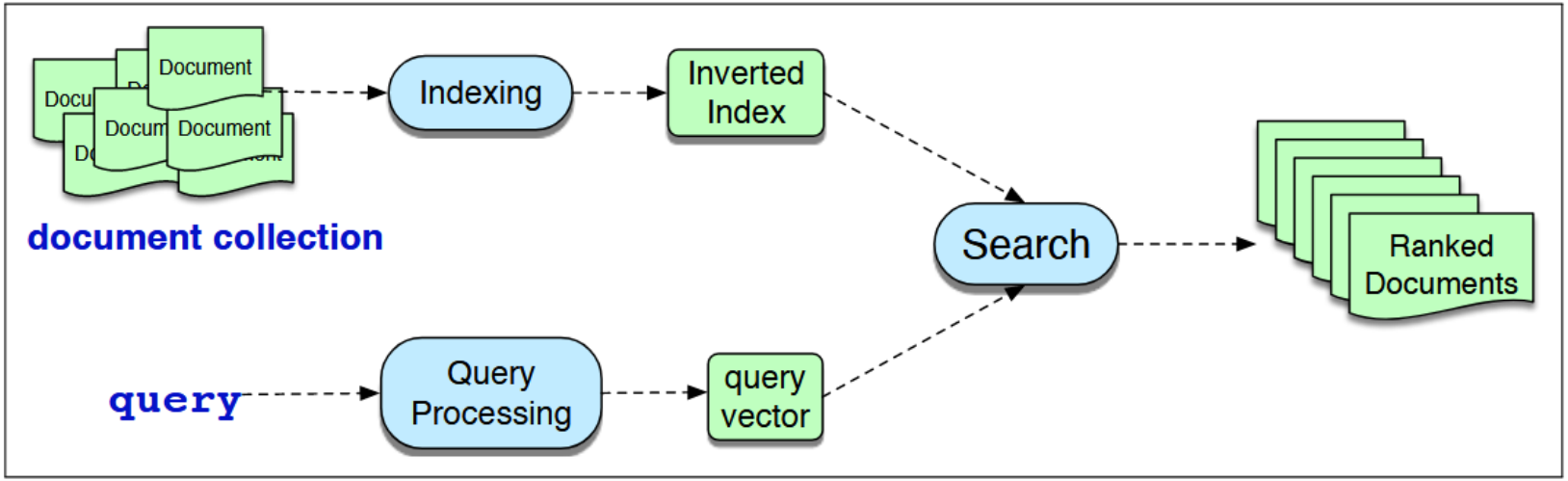
	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.074	0	0.22	0.28
good	0	0	0	0
fool	0.019	0.021	0.0036	0.0083
wit	0.049	0.044	0.018	0.022

BM25 transformations of bag-of-word vectors

- Modification of tf-idf
- Additional parameters:
  - *k* to control how much we care about word frequency
  - *b* to control how much we care about document length normalization
- Score of document *d* given query *q*:

$$\sum_{t \in q} \overbrace{\log\left(\frac{N}{df_t}\right)}^{\text{IDF}} \overbrace{\frac{tf_{t,d}}{k\left(1 - b + b\left(\frac{|d|}{|d_{\text{avg}}|}\right)\right) + tf_{t,d}}}^{\text{weighted tf}}$$

Traditional IR pipeline



- Return documents with most similar vectors to query vector (by cosine similarity)
- Inverted index: *term {document frequency} -> document_id1 [term frequency] document_id2 [term frequency]*
  - E.g. chicken {50} -> 774 [20] 32 [2]

Retrieval-augmented generation (RAG)

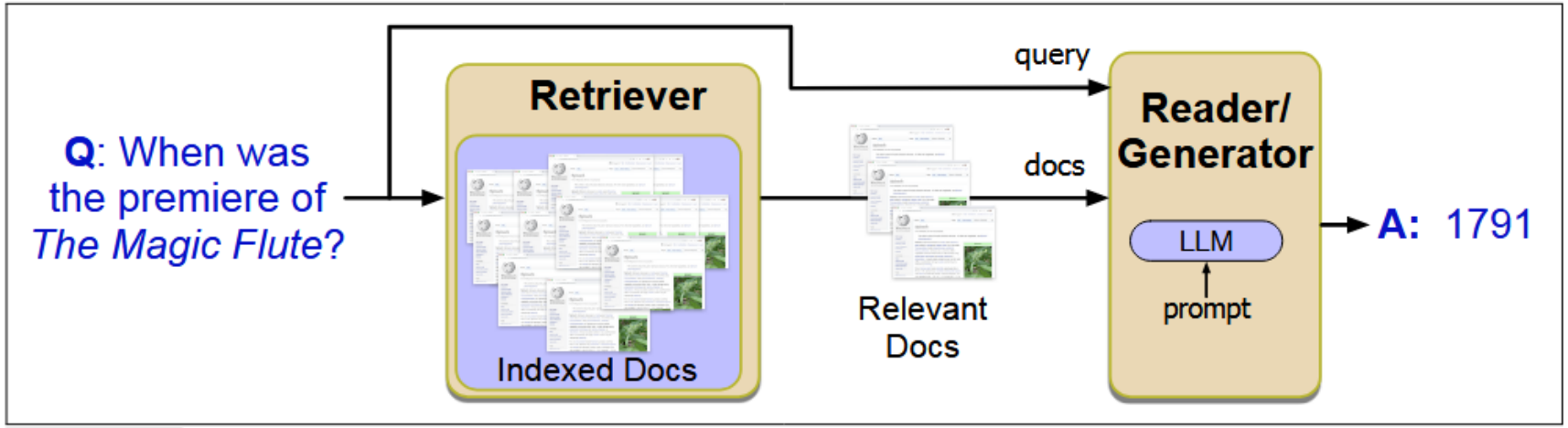
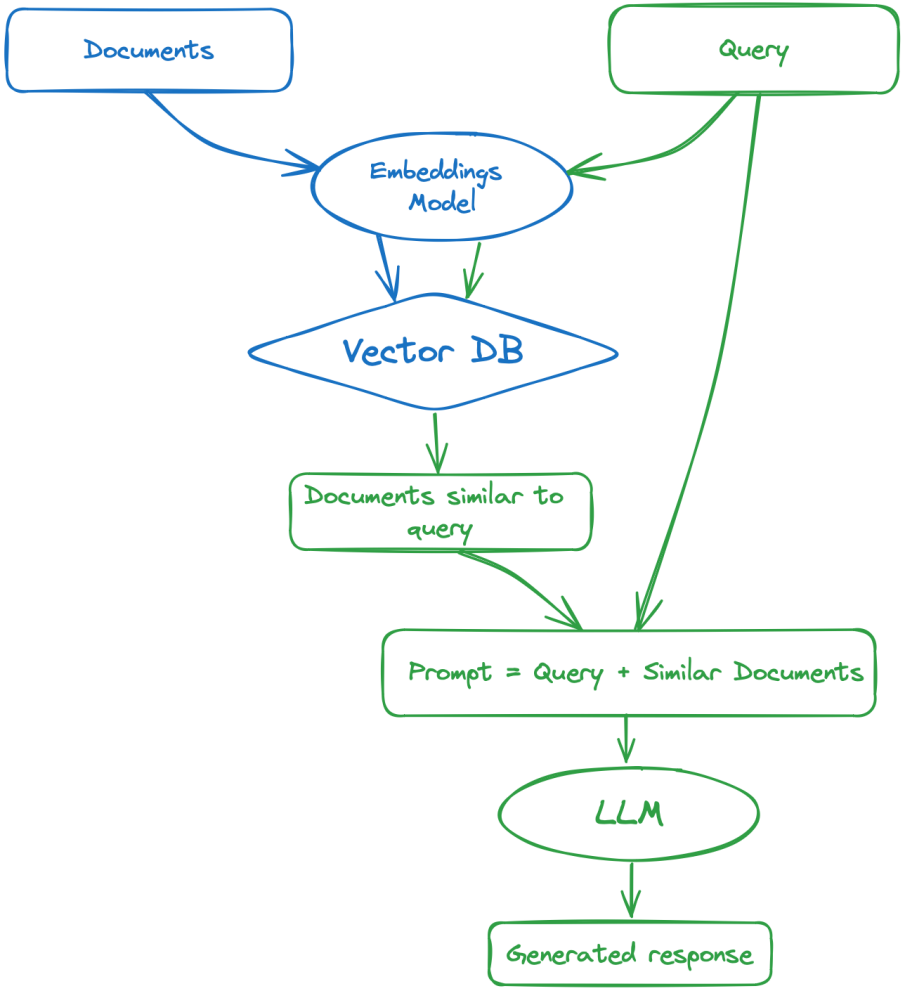


Figure 14.9 Retrieval-based question answering has two stages: **retrieval**, which returns relevant documents from the collection, and **reading**, in which an LLM **generates** answers given the documents as a prompt.

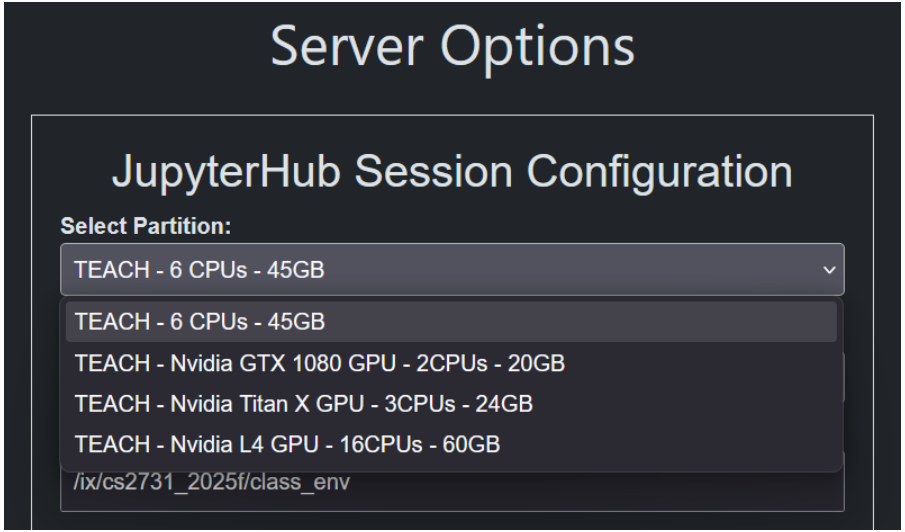


Credit: Maria Khalusova
https://huggingface.co/learn/cookbook/en/rag_zephyr_langchain

Coding activity

Notebooks: sparse IR and RAG

1. Go to this nbgitpuller link (also available on course website)
2. Start a server with **TEACH – 6 CPUs, 48 GB**
   - No need for GPUs
3. Load custom environment at **/ix1/cs1671-2026s/class_env**
4. Open **session25_sparse_ir.ipynb**
5. Open **session25_rag.ipynb**



Notebooks: what to record

- session25_sparse_ir.ipynb
  - Record:
    - Observations from trying different queries on MS MARCO
    - Mean reciprocal rank (MRR) on MS MARCO dev subset
- session25_rag.ipynb
  - Record:
    - Comparison between directly asking LLM and doing RAG
- If you finish early, try building a classic IR or RAG system on a new corpus of your choosing!

Wrapping up

- Classic information retrieval returns documents based on cosine similarity to the query's sparse embeddings, often transformed with tf-idf or BM25
- Retrieval-augmented generation provides relevant documents as context to an LLM to generate a response to prompts and questions
- Mean reciprocal rank (MRR) can be used for evaluation of information retrieval systems